

CLIENT SUCCESS STORY

Voomi Supply

Catalog Structuring for Marketplace Readiness

75%

FASTER PUBLISHING

70%

LOWER COST PER PRODUCT

300M+

LISTINGS MATCHED AGAINST

— BACKGROUND

The Catalog Problem in HVAC Distribution

Voomi Supply aggregates product catalogs from a distributed network of HVAC manufacturers and distributors and publishes them across marketplaces such as Amazon. The fundamental challenge: supplier catalogs are not designed for marketplace consumption.

SUPPLIER CATALOG REALITY

Catalogs arrive sparse, inconsistent, and structurally incomplete — built for internal use, not for the precision that marketplace matching demands.

MARKETPLACE REQUIREMENT

Publishing against 300M+ Amazon listings requires exact attribute alignment, compliance signals, validated imagery, and precise category classification.

THE CORE TENSION

“Marketplace performance is constrained not by supply, but by the structure of product data.”

— PROBLEM

Where the Publishing Workflow Broke Down

Existing Workflow

- Rule-based catalog matching
- Manual validation of product configuration
- Manual imagery and compliance review

Marketplace Risk Factors

- Incorrect matches lead to listing suppression
- Missing attributes reduce discoverability
- Compliance gaps create operational risk

CONSEQUENCES

- × Slow publishing cycles that delayed time-to-market for new suppliers
- × High operational cost driven by manual validation at every step
- × Limited scalability — supplier onboarding could not grow without headcount

— SOLUTION

Catalog Agents for Structured Marketplace Publishing

Paladio AI deployed **Catalog Agents** to construct a structured representation of each product *before* marketplace matching — replacing brute-force matching with a precision pipeline built on **catalog structuring**, **taxonomy alignment**, **compliance detection**, and **confidence-based automation**.

STEP 01

Catalog Structuring

- Extract and normalize product attributes
- Standardize naming conventions
- Construct structured product profiles

STEP 02

Taxonomy Alignment

- Classify products into consistent categories
- Improve downstream matching and discoverability

STEP 03

Compliance Signal Detection

- Identify hazmat classifications
- Flag multipack configurations
- Surface regulatory attributes automatically

STEP 04

Image Validation

- Automatically validate product imagery
- Reduce manual QA dependency

STEP 05

Targeted Matching Strategy

- Retrieve a constrained candidate set (~15 products per SKU)
- Apply structured attributes to improve precision
- Eliminate brute-force matching overhead

STEP 06

Confidence-Based Workflow

- Automate high-confidence matches end-to-end
- Route low-confidence cases to human validation
- Maintain full traceability of all decisions

— ARCHITECTURE

System Design Layers



Pre-Processing

Catalog normalization before any matching begins.



Constrained Matching

Targeted retrieval with structured attribute precision.



Confidence Scoring

Every match decision carries a scored confidence signal.



Human-in-the-Loop

Exceptions routed to review; automation handles the rest.

— RESULTS

MEASURABLE OUTCOMES

Speed, Cost, and Precision — Transformed Together

The Catalog Agent pipeline delivered what manual workflows could not: fast, accurate, scalable marketplace publishing without linear growth in operational cost.

75%

Reduction in product publishing time

70%

Reduction in cost per product published



Improved match precision and listing reliability

✓ Improved match precision and listing reliability across all SKUs

✓ Scalable supplier onboarding without linear headcount growth

✓ Automated compliance signal detection at scale

✓ Reduced operational bottlenecks and faster marketplace expansion

— IMPACT

From Manual Workflows to a Catalog Intelligence Layer

BEFORE

Manual, rule-based publishing workflows — slow, costly, and unable to scale with supplier growth



AFTER

A structured catalog intelligence layer that automates publishing, compliance, and matching at scale

THIS ENABLED

- Faster marketplace expansion across channels
- Improved compliance and listing stability
- Reduced operational bottlenecks at every stage

KEY INSIGHT

Marketplace performance is constrained not by supply,
but by the structure of product data.